

The Panel Loop

The nine agents in `.claude/agents/`, plus a tenth that reviews the panel itself — arranged around one constraint: how fast you can check the output.

Published as a repository · github.com/angsp0s/loop-engineering

Where this came from

I am building **Fade Readers** — an early-reading web app where children read real books *before* they know letter sounds. Every letter is drawn as the thing it starts with: the **a** is an apple, the **b** a bee, the **n** a ninja, each drawn inside the letter's own shape. A child who has never been taught a sound can still make something of **c a t**. Then the pictures fade — full picture, colored in, flat color, plain ink — until they are reading ordinary black type.

It is a small product with an unreasonable number of hard rules. Every letter has its own color, and no two may collide. Nothing may float outside a letter's footprint. Tapping a letter plays its *sound*, never its name, never “c is for cat”. One typeface for everything a child reads. A contrast floor that has to hold at every fade stage. No rewards, no streaks, no timers, no fast motion — ever. Builds are generated from templates and may never be hand-edited. And behind all of it, claims about children's reading that have to stay honest.

One person cannot check all of that on every iteration. So I built a review panel: nine expert-persona agents — a reading scientist, a curriculum lead, a children's interaction designer, an accessibility specialist, an art director, a responsible-design guardian, a children's-product lawyer, a marketer, an efficacy lead — that fan out in parallel and file findings before anything reaches me.

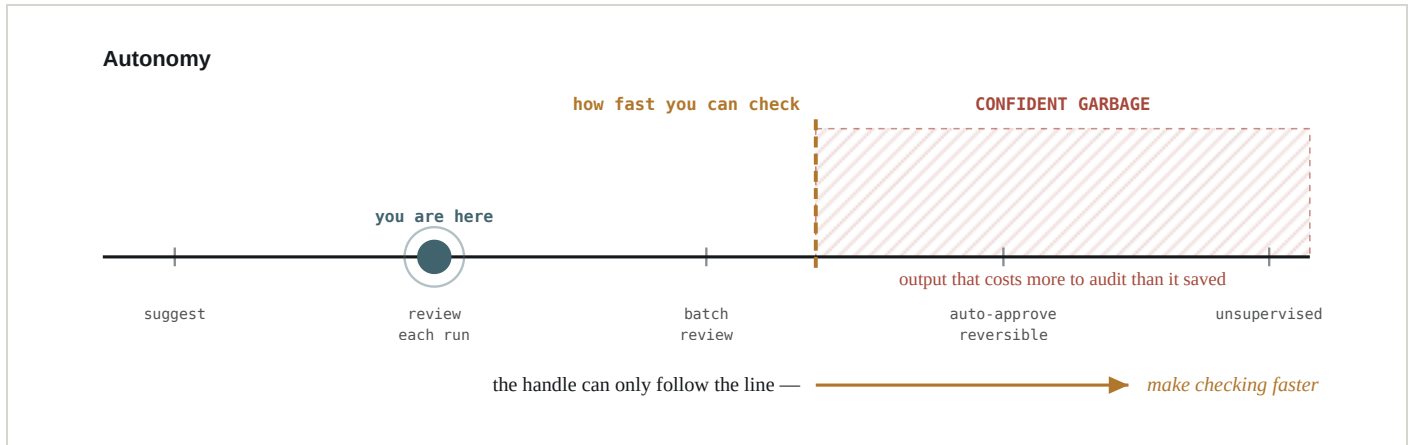
AND THEN THE PANEL HAD ITS OWN PROBLEM

Nine reviewers producing nine reports is not faster than reading carefully myself — it is **slower**, because now I have nine reports to read. The bottleneck had moved. It was never the reviewing; it was me, checking the reviewing.

Everything below is about that: not the app, but *the loop around the reviewing*.

Why the constraint is you

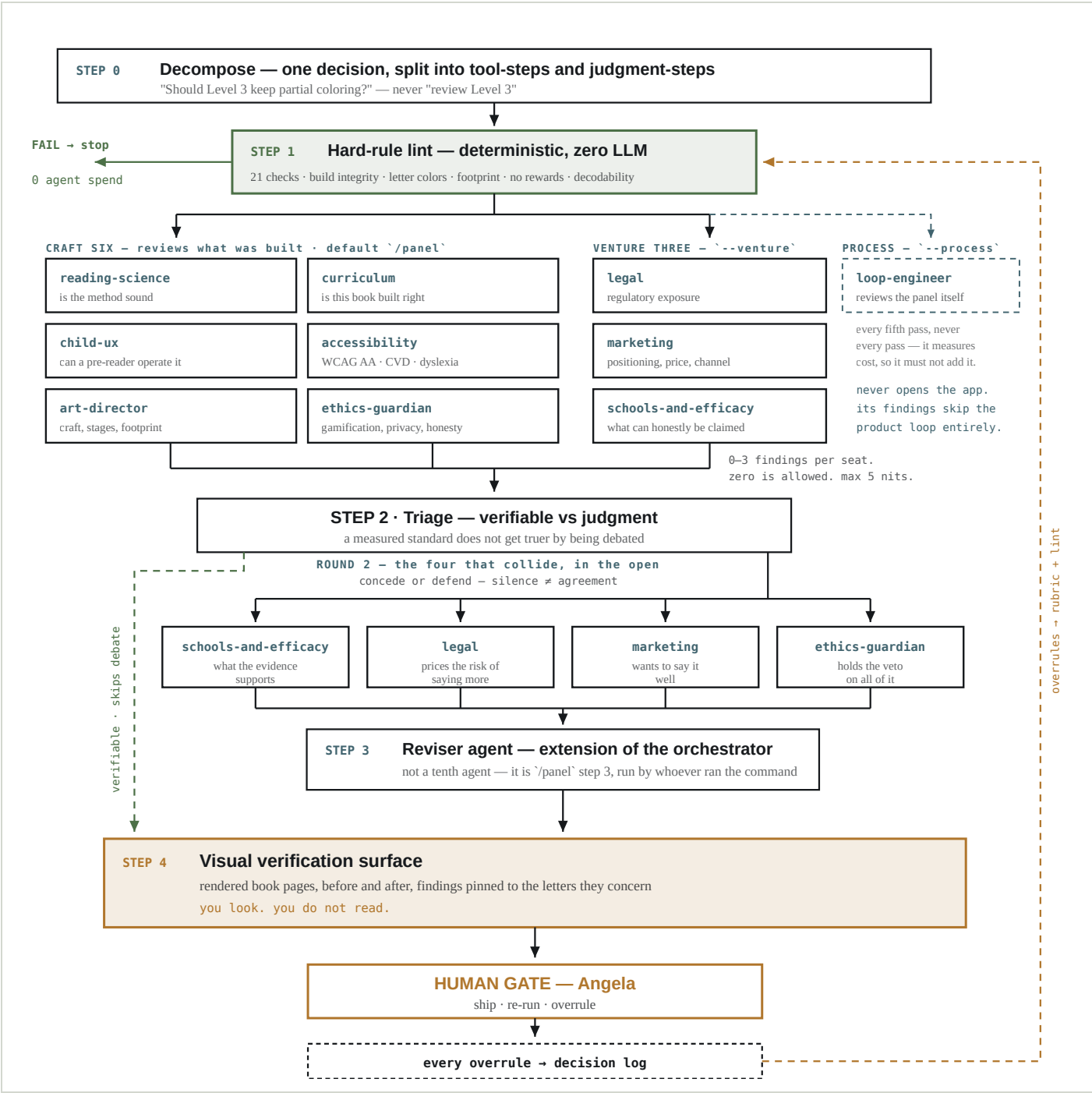
Andrej Karpathy – the generation–verification loop, the autonomy slider, "keep AI on the leash" ·
Software Is Changing (Again), June 2025



Karpathy's rule in one picture: autonomy is a slider you may move only as far as your verification reaches. Dragging the handle past the line doesn't buy speed — it buys output you can't check. Every build item below exists to push the ochre line rightward.

The loop

Cognition – share full traces, not fragments; conflicting implicit decisions · Don't Build Multi-Agents



Solid boxes run fixed checklists; dashed boxes explore freely — that difference is the autonomy dial. Two paths carry the argument: the green bypass, where a mechanically verifiable finding skips a debate it can't benefit from, and the ochre return, where disagreeing with the panel becomes rubric and lint rather than a feeling.

What you actually look at

Andrej Karpathy – a GUI uses "your computer vision GPU"; reading text is effortful · Software Is Changing (Again)

This is Step 4, drawn rather than described — because a page arguing for looking shouldn't hand you six paragraphs. One HTML page per panel run:

panel run · L3-partial-coloring · 08 Sep

LINT PASS ✓ Andika Bold ✓ letter colors ✓ footprint ✓ no reward markup ✓ template-built

NOW – LEVEL 2, TIER 2

PROPOSED – LEVEL 3, PLAIN INK

123

1 READING-SCIENCE

Tier 2 → plain ink in one step. Every other level moves one tier; this moves two. Is the fade withdrawing the cue faster than the tap-hint restores it?

2 CURRICULUM

Rubric P-04 satisfied — vowel remains distinguishable without color cue.

3 ART-DIR ETHICS

UNRESOLVED – shown, not settled

ART-DIRECTOR: plain ink is the whole promise of the fade. Ship it.

ETHICS-GUARDIAN: a child who stalls here taps to restore the picture. That is the crutch, not the safety net.

SHIP

RE-RUN

OVERRIDE

open full memo ↓

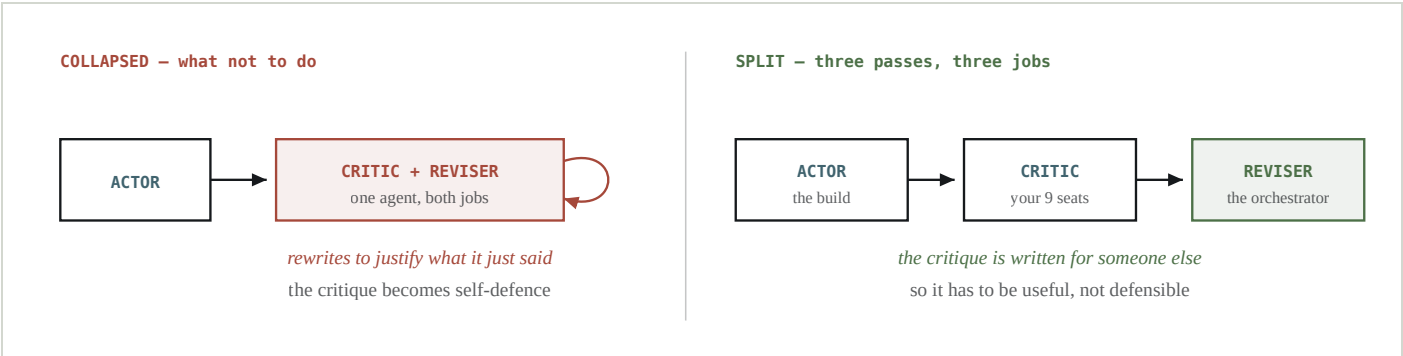
Mockup with example letters and colors — not your real palette. The prose memo still exists, one click down; it just stops being the first thing you touch.

WHY THIS SHAPE

Green strip means the hard rules held, so you skip straight to judgment. Findings are *pinned to the letters they concern*, so location is free information. And a live disagreement renders as both positions side by side rather than as the Reviser's summary of them — you see the argument, not a verdict about the argument.

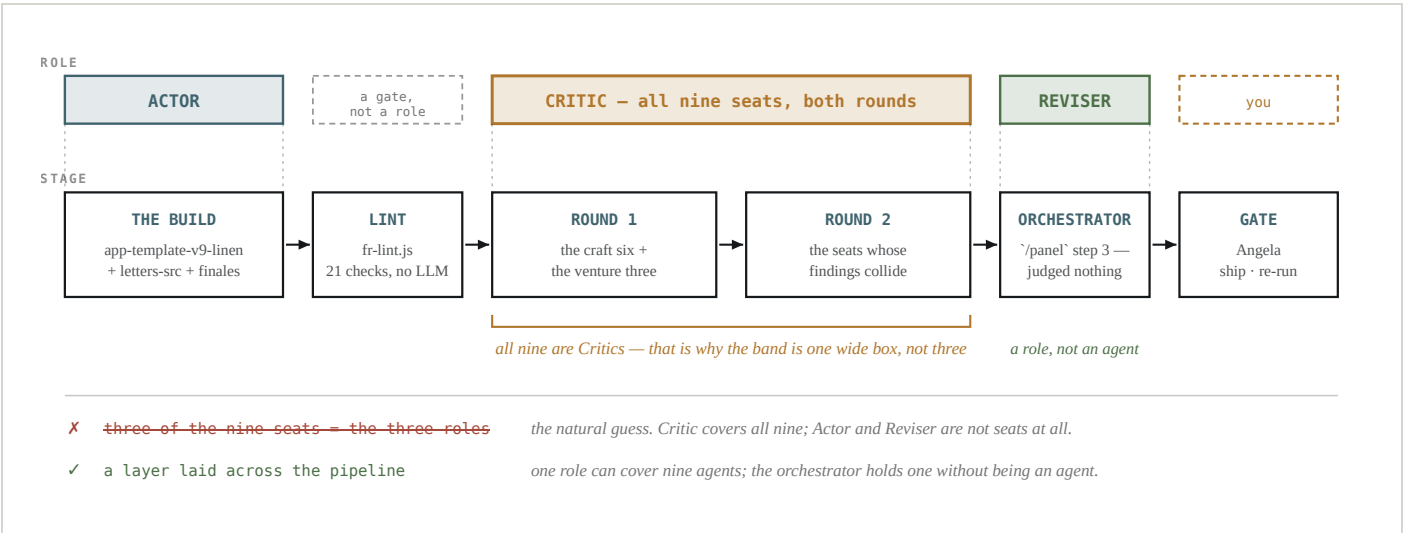
Actor · Critic · Reviser

Andrew Ng – the Reflection pattern · Agentic Design Patterns, Part 1, The Batch, 20 Mar 2024



The reason to split is not tidiness. An agent that critiques and rewrites in the same pass spends the rewrite defending its first opinion; separate them and the critique has to survive being handed to somebody who owes it nothing. No benchmark is offered here on purpose — nobody has isolated this split and measured it, and a number borrowed from adjacent work would read as evidence it hasn't earned.

Where Actor · Critic · Reviser lives in my workflow



The roles are not seats. **Actor** is the build — not an agent at all, just the files that made the work. **Critic** is all nine seats across both rounds; **accessibility** and **curriculum** leave after Round 1 because a measured standard doesn't improve by being argued over, not because they were something other than Critics. **Reviser** is not a tenth agent — it is `/panel`` step 3, performed by the orchestrator that launched the panel. It holds the role because it critiqued nothing, not because it is a separate entity. Call it the **reviser agent** — **an extension of the orchestrator**. The lint holds no role — it is a mechanical gate that runs before any of this. The rule underneath: no seat writes the consolidated report, because the moment **ethics-guardian** both argues to cut the payoff page and writes up what the panel concluded, the write-up becomes a defence of **ethics-guardian**. And **loop-engineer** holds no role in this picture at all — it reviews the picture.

UNTESTED — AND IT IS THE LEAST-WATCHED STEP IN THE LOOP

Nothing has ever checked the Reviser. Nine agents were interviewed, re-interviewed on their failed gaps, and audited mechanically. The consolidation step has had none of that — yet it is where `/panel` step 3 says "*merge duplicates, surface disagreements under their own heading, do not soften, drop the noise.*" Every one of those is a place a blocker can quietly become a concern, or a live disagreement can get resolved silently instead of being put to you in two lines.

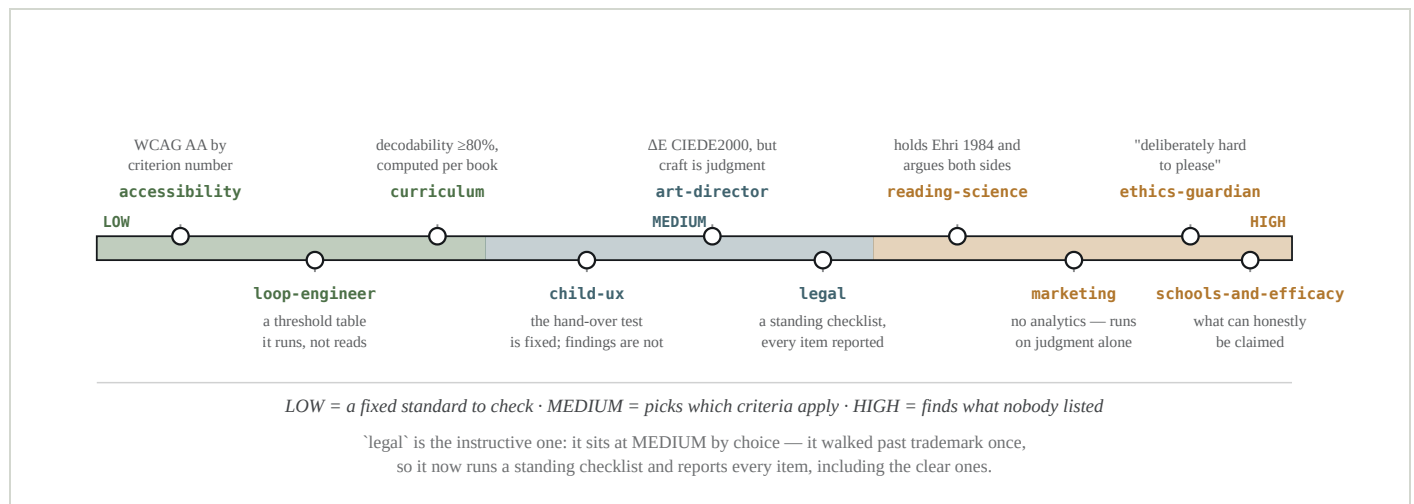
It is also the step with the worst incentive: the orchestrator that consolidates is the same one that chose the question, launched the panel and will present the result. **Merging is where a run can be made to look better than it was.**

What would test it: take a past pass with a known disagreement, consolidate it twice, and check that the disagreement survives to the top of both. That is now `loop-engineer`'s to measure — a Round 1 blocker that appears as a concern in the consolidated report, or a disagreement that appears only in one seat's section and not under its own heading.

Autonomy, dialed per seat

Andrew Ng — "different degrees to which systems can be agentic" · The Batch 253, 12 Jun 2024

The dial is not something to bolt on — it is *already latent in the agent files*. Some seats carry a fixed standard they check against; some exist to find what nobody wrote down. Reading the ten files, this is where each already sits.



Nothing here was assigned; it was read off the files. *accessibility* cites criteria by number, *curriculum* computes decodability per book, *art-director* measures ΔE — those are checkable standards. *ethics-guardian* is told in its own file to be the least agreeable reviewer, and *marketing* works with no analytics at all, so neither can be run from a list. The dial is a description, not a policy.

The tenth seat — loop-engineer

The nine ask whether Fade Readers is any good. This one asks whether the *reviewing* is any good, and whether it earns its cost. Nothing else was looking at that — and one day's audit made the case on its own.

CAUGHT BY NOTHING Three stale specs in a day The payoff-art count, the curriculum ladder, the octopus color. Each individually findable. Nobody's job was to notice they were a <i>pattern</i>.	CAUGHT BY NOTHING A number that was guessed A reviewer asked for a count answered "approximately 48" against an actual 59 — while every quotation it gave was exact.
CAUGHT BY NOTHING A fix pass that broke things Repairing nine findings introduced two new ones. Only the re-run caught it. No product reviewer would ever file that.	

WHAT IT MEASURES — THRESHOLDS, NOT VIBES

Evidence compliance under 90% · a seat filing the same non-zero count every pass · a Round 1 finding vanishing from Round 2 without an explicit concede · any spec claim the code contradicts · any number not traceable to a command · a re-raise missing the required "*rejected on <date> because <reason>; re-raising because <what changed>*" line · your gate agreement rate · report length against findings that changed a decision · - -all run on a single-domain question.

TWO GUARDRAILS, AND THEY ARE THE DESIGN

It never opens the app. Its inputs are `wiki/feedback/panel/`, `lint-report.json`, the agent files and the decision log. A process reviewer with opinions about letter colors has become a tenth product reviewer, which is worse than not existing.

It runs every fifth pass, not every pass. It measures cost; adding cost to every run would make it the thing it is checking. Most of its passes should end in "nothing found" — and a clean process pass is exactly what tells you the loop is safe to lean on.

It carries principles, not people. Verification bounds autonomy · compute, never estimate · decompose before adding autonomy · evals over intuition · read parallelises, writing collides. Deliberately unattributed: a reviewer that invents a position for a living author has done the exact thing this seat exists to catch.

Who gets what

Ten seats, so the packet is scoped by group rather than named per agent. Everything below already lives in `_PROTOCOL.md` §1, which every reviewer is required to read first — one file to keep current instead of ten.

PACKET ITEM	CRAFT SIX	VENTURE THREE	LOOP-ENGINEER	WHY
the_question	●	●	●	One decision with a default attached. Batch size is bounded by what you can verify.
CLAUDE.md	●	●	●	Grounding step 1. Three of its facts went stale in a single day — the loop engineer diffs it against the code rather than trusting it.
lint-report.json	●	◐	●	A rule the lint covers and passes is settled, not a finding. A failing report stops the pass.
letters.json	●	○	○	Cite the manifest; never re-derive color, word or tier by parsing <code>letters-src.js</code> .
feedback-log + prior panel reports	●	●	●	A finding Angela rejected is dead unless the re-raise names the ruling, the date and what changed.
the vault	●	◐	◐	115 notes. Scoped by role — the loop engineer reads <code>wiki/feedback/panel/</code> and the agent files, not the letters.
the built app	●	◐	○	The loop engineer never opens it. A process reviewer with opinions about letter colors is a tenth product reviewer.
your_raw_text	◐	●	●	The high-autonomy seats need your intent to argue against it; a seat running a fixed standard is only anchored by it.
your_leaning	○	○	○	Never. A panel that knows the CEO's preference stops being a panel — and §5c already tells them to challenge you.

● gets it ◐ scoped subset ○ not needed for the role ○ never, by design

```
letters.json    // emitted by fr-lint.js on every run, never hand-written
{ "b": { "word": "bee", "color": "#f8d67c", "cBase": "#b09033",
        "contrastOnCanvas": 3.0, "survivesFade1": false, "font": "Andika-Bold" } }
```

This file is what "build for agents as a third consumer" means concretely. Humans get a GUI, computers get an API, and agents get neither — they're machine-speed but read like people. A reviewer checking whether **b** renders at the right tier shouldn't have to infer it from CSS.

Evals — the unglamorous version

Andrew Ng – evals and error analysis as the biggest predictor of progress · The Batch, 15 Oct 2025

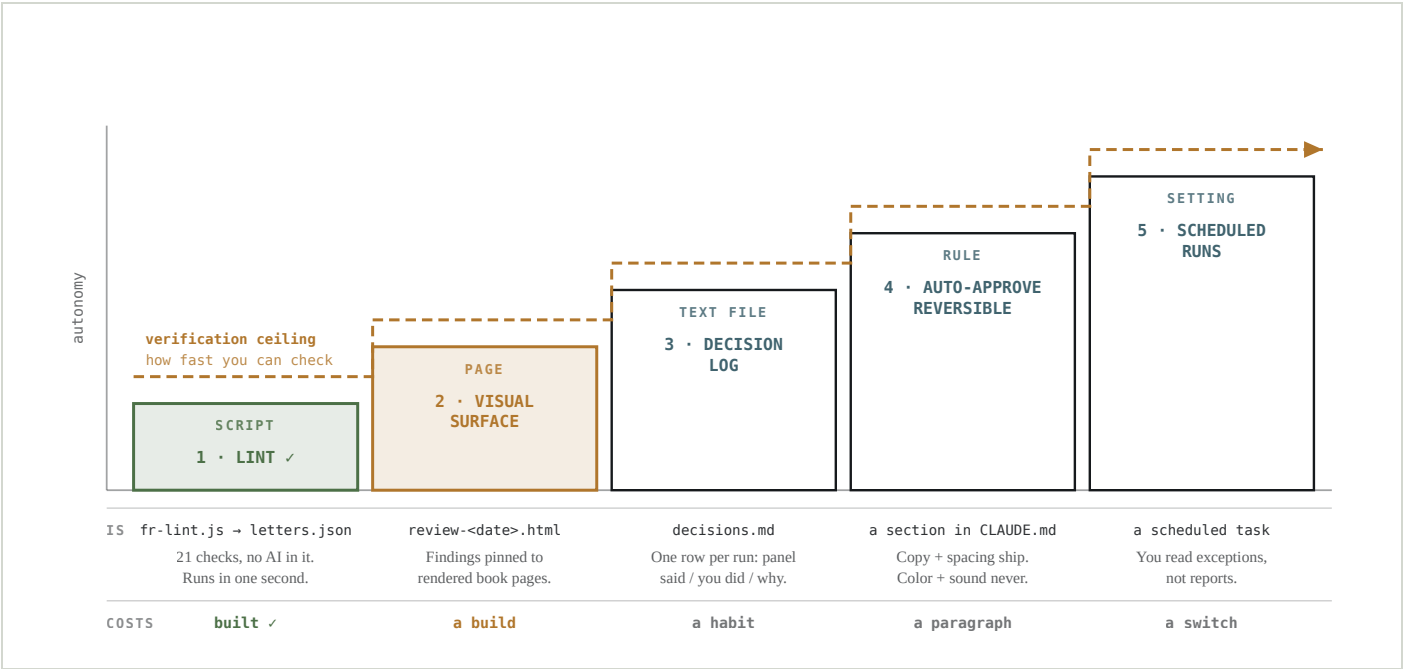
Ng calls disciplined evaluation the single biggest predictor of whether an agentic system works. It doesn't start as infrastructure. It starts as a log.

FROM RUN 1 Log Every gate decision: what the panel recommended, what you decided, one line on why they differed.	~ RUN 10 Count Tally the recurring types. “ethics-guardian over-cuts.” “child-ux misses bedtime context.” Objective counting before any scoring.	ONGOING Fix Each recurring type becomes a rubric line, a lint rule, or a dial change. Designed out, not caught again.	ONCE STABLE Judge LLM-as-judge scores findings against the rubric, so low-value findings never reach your surface at all.
--	--	---	---

Your agreement rate with the panel is the number that unlocks the ladder below. Not a feeling that it's working — a rate you can point at.

The autonomy ladder

Andrej Karpathy – autonomy moves only as far as verification reaches • Software Is Changing (Again)



The same slider from the top of the page, unrolled over time. Each rung is one concrete thing — a script, a page, a file, a paragraph, a setting — and the bottom row is the part that surprises people: only rungs 1 and 2 are real building. The dashed line is how fast you can check; a block may only rise as high as the line has already reached. Build rung 4 before rung 3 and the block sticks out above it, which is the definition of confident garbage.

STATUS

Rung 1 is built. `fr-lint.js` sits in the project folder — 21 deterministic checks across build integrity, letter rules, sound, values, decodability and fonts, plus it emits `letters.json` on every run. First run caught a stale build: `finales-src.js` was edited after the last rebuild, so the shipped `beta/index.html` carried a payoff illustration whose dog clipped the page frame — for a day. Rebuilt and shipped 2026-09-08. Rung 2 is next.

Sources

- [Karpathy — Software Is Changing \(Again\)](#), YC AI Startup School, June 2025 (full transcript) · [summary](#)
- [Ng — Agentic Design Patterns Part 1](#), The Batch, 20 Mar 2024
- [Ng — letter on degrees of agency](#), The Batch 253, 12 Jun 2024
- [Ng — Evals and Error Analysis, Part 1](#), The Batch, 15 Oct 2025
- [Cognition — Don't Build Multi-Agents](#)
- [LangChain — How and when to build multi-agent systems](#)